The Value of Auditor Industry Specialization: Evidence from a Structural Model

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Abstract

This study investigates the value of auditor industry specialization. In the first step, we use a discrete choice model to derive the first-order demand for auditor industry specialization. Our results reveal that clients have a general preference for auditor industry specialization, relating to both audit firm and audit office specialization. Further, we observe that specializations at the audit firm and audit office level are substitutes. We also find that larger and more complex clients have a stronger demand for industry specialization at the audit office level. In the second step, we use the results from the discrete choice model to quantify the value of auditor industry specialists for clients. We find that the overall value of industry specialization aggregated across all clients is 5.2 million US Dollars (0.36% of audit fees) and that industry specialization at the firm (office) level is decisive for auditor choice in 4% (6%) of all cases.

JEL Codes: M42, M48, L11, L84

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1 Introduction

It is a fundamental question whether auditing adds value for clients. However, identifying this value is challenging in settings where auditing is mandatory (DeFond and Zhang, 2014). Prior studies rely on voluntary audit settings and focus on the potential economic benefits of hiring an auditor, e.g., lower costs of debt and higher credit ratings (e.g., Lennox and Pittman, 2011). In our study, we use the discrete choice approach recently introduced to auditing (Gerakos and Syverson, 2015) to quantify the value of hiring an auditor with industry specialization within a mandatory setting. Specifically, we structurally estimate clients preference over this auditor attribute and quantify its monetary value in counterfactual analyses based on the utility loss clients suffer when being restricted from hiring industry specialists.

It is an unresolved empirical question whether auditor industry specialization provides a positive net value for clients. Clients may profit from the insights of an auditor with deep industry knowledge and an understanding of industry-specific accounting issues (e.g., Dunn and Mayhew, 2004). Furthermore, industry specialists may enhance the quality and the credibility of the financial reports (e.g., Balsam et al., 2003, Reichelt and Wang, 2010), allowing clients to signal the credibility of their financial reporting to the capital market by hiring an industry specialist (e.g., Datar et al., 1991). However, some more recent studies suggest that the impact of industry specialists on audit quality is limited (e.g., Minutti-Meza, 2013, Jiang et al., 2019) or even detrimental in times of heightened industry-specific risk (Cassell et al., 2019). Furthermore, clients may be reluctant to hire an industry specialist serving many similar clients due to the risk of information spillovers (e.g., Aobdia, 2015; Bills et al., 2020).

This study uses a discrete choice model to investigate the first-order demand for industry specialization and to quantify its value. In our context, first-order demand means that the average client derives utility from hiring an industry specialist.[1] In contrast, the second-order effect means that some client types, e.g., larger or more complex clients, derive a higher utility from hiring an industry specialist than others.[2] Only knowledge about the

[1] Prior research does not provide conclusive evidence on the first-order demand for industry specialization. Prior studies on audit fees can be categorized as investigating first-order effects, e.g., whether clients are, on average, willing to pay a fee premium for auditor industry specialization (e.g., Francis et al., 2005). However, it is challenging for these studies, using the traditional regression framework, to disentangle supply and demand effects (for further discussion on this topic, see Gerakos and Syverson, 2017).

[2] Prior studies on auditor choice typically focus on second-order effects. These studies investigate whether clients with more severe agency problems or clients with a stronger audit committee are more likely to hire an industry specialist (e.g., Godfrey and Hamilton, 2005; Ettredge et al., 2009; Srinidhi et al., 2014; Lai et

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first-order demand reveals whether auditor industry specialization provides, on average, a net benefit for clients. This knowledge provides insights on how much industry specialization contributes to the differentiation of audit services and, thus, the market power of audit firms. It helps audit firms to assess whether they can use industry specialization to signal their quality to clients and how much they should invest in developing it (GAO, 2008; Minutti-Meza, 2013). It informs clients how important industry specialization typically is for auditor selection in practice.

Discrete choice models are suitable for investigating the first-order demand for individual attributes of specific goods or services (e.g., Petrin, 2002; Chu and Chintagunta, 2009, 2011). Accordingly, they are widely used in research in industrial organization (e.g., Berry et al., 1995) and quantitative marketing (e.g., Dubé et al., 2002). One advantage is that they can accommodate rich client heterogeneity. Gerakos and Syverson (2015) apply the discrete choice models to auditing, expressing the demand for the audit service of a specific audit firm based on the attributes of the audit firm and the match between characteristics of the audit firm and the client. We extend their model by including industry specialization as an explanatory variable in the model. We include this attribute at both the audit firm and audit office level, helping us to disentangle these two drivers of auditor demand. Furthermore, including multiple auditor attributes makes it possible to separate industry specialization from other auditor attributes and to investigate interdependencies.

The discrete choice approach provides a framework for treating the potential endogeneity of audit fees seriously. Audit fees are endogenous to audit demand if they are related to some auditor attributes that are not perfectly observable for the audit researcher, e.g., audit quality or auditor reputation. We use a control function approach to address this issue (Petrin and Train, 2010; Wooldridge, 2015). The basic idea is to derive a variable that, when included in the empirical model, renders the variation of the endogenous variable exogenous (Heckman and Robb, 1985). Applied to our setting, the residual of an audit fee regression reveals unobserved demand factors and, thus, we include it in the demand estimation to correct for the endogeneity of audit fees.

One key advantage of the discrete choice model is that it offers the opportunity to perform policy evaluations. Applying the random utility maximization approach rooted in McFadden's choice theory (McFadden, 1973), it is possible to use the parameters obtained in the demand estimation to predict clients' auditor choices in a counterfactual setting.

al., 2017).

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Gerakos and Syverson (2015) employ this technique to evaluate the losses in consumer surplus associated with an audit firm rotation requirement and a sudden market exit by one of the Big 4 audit firms. Guo et al. (2017) use a similar approach to estimate the effects of introducing a joint audit requirement on audit market structure and consumer surplus. In a recent working paper, Cheynel and Zhou (2020) use a dynamic discrete choice model to investigate the consequences of mandatory auditor rotation for audit quality.

Using the US audit market as a setting, we find that clients have a first-order demand for auditor industry specialization both at the audit office and the audit firm level. As we control for a range of variables including auditor attributes, our findings show that clients perceive industry specialization as a distinct positive auditor attribute beyond auditor brand name. Furthermore, we test for interdependencies between auditor attributes, detecting a substitutive relation between specialization at the office and the firm level. We also find that industry specialization at the office level substitutes for audit office size and Big 4 status. Finally, we employ the discrete choice model to revisit the findings of prior research that companies with more severe agency problems are more likely to hire industry specialists (e.g., Godfrey and Hamilton, 2005; Ettredge et al., 2009; Kang, 2014). Consistent with prior findings, we observe that larger and more complex clients prefer industry specialists. As a new insight, our findings show that these second-order effects are only present at the audit office but not at the audit firm level.

Next, we perform counterfactual analyses to assess the economic importance of auditor industry specialization. In a first step, we estimate the value of industry specialization based on the loss in consumer surplus that clients suffer when they are restricted from hiring auditor industry specialists. Aggregating across all clients, we find that the value of auditor industry specialization is 5.2 million US Dollars per year (0.36% of audit fees), thereof 1.9 million US Dollars at the audit firm level and 3.3 million US Dollars at the audit office level. These values are substantially smaller than our estimated average value of auditor brand name for each of the Big 4 audit firms, which is 354.2 million US Dollars (25.0% of audit fees). In a second step, we investigate the impact of auditor industry specialization on the audit market. We find that a limited number of clients would switch to another audit firm if the value of industry specialization ceased to exist. In such a scenario, 6.03% (4.34%) of clients originally hiring an office (firm) specialist would switch to another audit firm. Finally, we find that audit firms would gain up to 10% market share in terms of the number of clients if they became specialists in all industries. Non-Big 4 audit firms would

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profit especially strongly in this scenario by attracting larger clients, thereby increasing their market share in terms of clients' assets substantially.

Given that prior research on auditor demand regularly focuses on industry specialization (DeFond and Zhang, 2014), it might be surprising that the value of this attribute is much smaller than the value of the auditor brand name. However, it is consistent with the notion that the net benefit of hiring an industry specialist is unclear. One concern could be that our estimates for the value of auditor industry specialization are downward biased by the auditor brand effect. Such a bias could be present if these two auditor attributes were highly correlated and difficult to disentangle. We address this concern in two ways. First, we check for the correlation between auditor industry specialization and auditor brand name. Overall, we find that the correlation is weak.[3] Second, we observe that our dataset offers rich variation of auditor industry specialization status for each Big 4 audit firm across industries, regions, and over time. This variation allows us to disentangle auditor industry specialization as a time-variant factor from the time-invariant auditor fixed effects.[4]

Our paper provides several contributions. First, it expands research on the demand for auditor industry specialization. Our study provides novel evidence on the first-order demand, revisits findings on the second-order demand for auditor industry specialization, and explores interdependencies. Second, our study addresses the fundamental question of whether auditing adds value within the mandatory audit setting. We focus on the incremental value of auditor industry specialization as a key auditor attribute. Third, our study provides new paths to future research in auditing by showing how the discrete choice approach can be applied to study fundamental issues in auditing from an alternative angle.

Our study provides policy implications. First, one key concern of regulators is audit market concentration. In this context, regulators wonder how much industry specialization matters for auditor selection (GAO, 2008) and whether industry specialization is a key capability smaller audit firms need to develop to compete with larger audit firms (EU, 2014). Our study on the first-order demand for industry specialization provides novel insights on these topics. Second, our study helps to evaluate unintended consequences of regulatory actions, e.g., requiring auditor rotation or breaking up the Big 4 audit firms. Auditor

[3]The correlation between audit firm fixed effects and auditor industry specialization at the firm level (office level) ranges from -0.06 to +0.38 (+0.13 to +0.28) (see Online Appendix Table A1).

[4]We identify only one SIC2 industry lacking this variation because an audit firm is an industry specialist for the full sample period at the audit firm and office level (see Online Appendix Table A2 and Table A3). We perform a robustness check dropping this industry and observe that our results remain robust (see Online Appendix Table A4).

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rotation may result in substantial consumer surplus losses because it forces clients to switch away from their most preferred auditor (Gerakos and Syverson, 2015). Our study reveals how much of this surplus loss is attributable to auditor industry specialization relative to auditor brand name. Breaking-up audit firms, as considered in the UK (CMA, 2019), could make it more difficult to satisfy the need of Big 4 clients for specialized industry expertise (FT, 2018). Our study uncovers how much clients would lose if audit firms lose industry specialization in this scenario.

The remainder of the paper is organized as follows. Section 2 discusses techniques for estimating auditor demand. Section 3 describes our method. Section 4 provides the results of the demand estimation. Section 5 presents the counterfactual analyses for quantifying the value of industry specialization. Section 6 provides the conclusion.

2 Application of the discrete choice approach for estimating audit demand

2.1 Opportunities

The discrete choice approach provides the opportunity to address some of the constraints faced by prior research. Prior studies on the demand for industry specialization use the traditional regression framework (e.g., Godfrey and Hamilton, 2005; Ettredge et al., 2009; Srinidhi et al., 2014; Lai et al., 2017). These studies typically employ a logit model with industry specialization of the selected auditor as a binary dependent variable and client characteristics reflecting size, riskiness, and profitability as independent variables. This binary choice model has several constraints. First, it is restricted to investigating second-order demand effects. That is, it can only identify whether some clients are more likely to hire an industry specialist than others but not whether clients, on average, have a demand for industry specialization. Second, the use of industry specialization as a dependent variable means that it is not possible to include industry specialization at the audit firm and audit office level simultaneously, making it impossible to disentangle both factors. Third, the use of a binary dependent variable makes it infeasible to control for the observable characteristics of each of the Big 4 audit firms available for selection, triggering the risk that the results may be confounded by auditor brand name and other auditor attributes including the audit

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firms' pricing strategy.[5] Relatedly, it is also not possible to test for interdependencies between auditor industry specialization and other auditor attributes.

Our structural application of the discrete choice approach addresses these shortcomings of prior studies, enabling us to provide novel evidence on the value of industry specialization. We use the discrete choice approach to estimate for each client the conditional probability of choosing a specific audit firm based on a utility framework. Specifically, using the specific audit firm instead of industry specialization as the dependent variable makes it feasible to include a rich set of audit firm attributes as independent variables. First, we include industry specialization as a main effect to investigate the first-order demand for industry specialization. Second, we insert industry specialization at both the audit firm and the audit office level simultaneously, enabling us to disentangle whether specializations at both levels matter and whether interdependencies across these two levels exist. Third, we add multiple auditor attributes, making it feasible to separate industry specialization from other auditor attributes and to investigate interdependency among auditor attributes. In particular, we include auditor fixed effects to control for auditor brand name. Furthermore, the discrete choice approach accommodates for rich client heterogeneity and makes it possible to integrate audit fees in a manner that treats endogeneity concerns seriously.

One key advantage is that the discrete choice model as a structural approach makes it possible to perform policy evaluations.[6] Applying the random utility maximization approach rooted in McFadden's choice theory (McFadden, 1973), it is possible to use the parameters obtained in the demand estimation to predict client choices in a counterfactual scenario. Using this approach, we investigate how industry specialization shapes the audit market structure and we quantify the value of industry specialization for clients in monetary terms.

[5] One key challenge is to disentangle the demand for auditor industry specialization from the demand for auditor brand name. Prior studies on auditor choice attempt to do so by restricting their sample to the Big 4 audit firms (e.g., Abbott and Parker, 2000; Ettredge et al., 2009). However, while this approach handles the issue of potential brand name differences between the Big 4 and non-Big 4 audit firms, it implicitly assumes that there is no differentiation in auditor brand name within the Big 4 audit firms. Recent audit research calls this assumption into question (e.g., Francis et al., 2014; Brown and Knechel, 2016).

[6] The discrete choice model makes it possible to use a structural approach considering both the benefits and costs associated with choices. As pointed out by Gow et al. (2016), structural approaches are valuable tools for addressing institutionally rich problems and for estimating structural parameters to provide interesting counterfactuals. Gow et al. (2016) also state that the use of structural models in accounting research has been limited and that applying structural approaches to accounting research questions would be very useful. Recent studies applying a structural approach to accounting issues investigate disclosure issues (e.g., Cheynel and Liu-Watts, 2019; Bertomeu et al., 2020; Zhou, 2021), earning management (e.g., Gerakos and Kovrijnykh, 2013; Zakolyukina, 2018; Bird et al., 2019), CEO compensation (e.g., Gayle et al., 2020; Li, 2021) and investment decisions (Breuer and Windisch, 2019; Choi, 2021).

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2.2 Assumptions and limitations

The discrete choice approach has limitations. First, it requires assumptions about economic behavior. We assume that clients choose their auditor each year in a static framework. This framework does not fully integrate the dynamic features of auditor choice, i.e., the utility clients obtain from rehiring the auditor in future periods (Cheynel and Zhou, 2020). Nevertheless, our framework enables us to capture some dynamic components of the choice setting, e.g., the utility from rehiring the prior year auditor. Further, Gerakos and Syverson (2015) show that a static model can well describe clients' actual auditor choices.

Second, our implementation of the discrete choice approach focuses on the demand side. The limitation is a neglect of the supply side. For example, we assume that audit firms are not capacity constrained and are able and willing to submit a tender for all audit engagements. We consider the supply-side in a simplified manner by modeling the pricing function of each audit firm. We do so without imposing any specific type of competition. We simply assume that observed prices reflect equilibrium outcomes and that the pricing function is stable over time and common knowledge to all parties involved.

Third, the implementation of the discrete choice approach as a logit model imposes the independence of irrelevant alternatives (IIA) property. According to this property, introducing or removing additional choices should not change the relative probability of choosing x over y. This property is sometimes unrealistic as described in the classic "Red Bus, Blue Bus" example. We address this issue by including a number of observable client characteristics in interaction with audit firm fixed effects into the utility model, thereby allowing for rich substitution patterns (Ackerberg et al., 2007, p. 4185). Using a similarly specified model, Gerakos and Syverson (2015) find that removing the option of Arthur Andersen does not have a large effect on the relative probability of choosing one Big 4 audit firm over another Big 4 audit firm. This finding suggests that the IIA property is not a major concern in the audit setting when allowing for rich substitution patterns.

Fourth, the discrete choice approach estimates conditional choice probabilities. Applied to our setting, this approach means that for each client the preferences for one auditor are identified relative to the other auditors in the choice set. In other words, client characteristics do not vary across different auditor choices. Accordingly, this approach does not reveal the absolute level of client utility. Further, it is not possible to identify direct effects of client characteristics on the absolute level of utility obtained from an audit, e.g., whether larger clients obtain a higher utility from an audit than smaller clients. However, it is possible to

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derive the incremental utility of specific auditor attributes and the match between client and auditor characteristics, e.g., whether larger clients obtain higher utility from hiring an industry specialist than smaller clients.

Fifth, the discrete choice approach in our paper is a structural model. Bertomeu et al. (2021) provide a concise discussion of the advantages and disadvantages of structural models. They argue that the strong economic assumptions imposed in structural models provide opportunities for new insights but carry the cost of higher complexity. Therefore, structural models shall be considered as complementary to the results of studies using the traditional regression framework. The structural application of the discrete choice approach used in our study relies on the assumption that clients maximize their utility and that their choices reveal their preferences. It is flexible in specifying clients' preferences over auditor choices without being restrictive about specific economic tradeoffs driving these preferences.

3 Method

3.1 Demand model

We use the discrete choice model to estimate the demand for auditor industry specialists. Applying the random utility maximization approach rooted in McFadden's choice theory (McFadden, 1973), we assume that clients choose the audit firm that provides the highest net benefit. In our context, the net benefit captures the match between client characteristics and auditor type, auditor attributes, auditor tenure, and audit fees. Consistent with prior literature, we assume that the different components of the net benefit can be aggregated by adding them up (e.g., Berry et al., 1995; Dubé et al., 2002; Train, 2009). Each client can choose from a choice set including the Big 4 audit firms (EY, Deloitte, KPMG, and PwC), and the group of all other small audit firms. Following Gerakos and Syverson (2015), we group the small audit firms into one category and define them as the outside option, assuming that they provide a homogenous level of audit service.

The deterministic part V_{ijt} of the utility of client i in period t from choosing one of the Big 4 audit firms and small audit firms, $j = 0, \ldots, 4$ is given by:

$$V_{ijt} = \alpha_0 ISP_{ijt} + \alpha_1 \chi_{ijt} + \sum_{k=1}^{4} (\beta_{1k}\delta_k + \beta_{2k}\delta_k \tau_{it}) - \alpha_2 \ln(p_{ijt}) + \xi_{jt} \tag{1}$$

Variable ISP_{ijt} denotes auditor j's attributes: being an industry specialist in the industry

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of the client i based on the SIC2-industry classification.[7] Variable χ_{ijt} indicates the tenure between client i audit firm j in period t. Variable δ_k is the dummy variable for each of the Big 4 audit firms; parameter β_{1k} captures the auditor fixed effect that represents the mean utility for all clients choosing auditor k.[8] We interact the dummy variable for each auditor with client characteristics τ_{it} to capture the heterogeneous preferences of clients, e.g., the preference of larger clients for larger audit firms. Variable $\ln(p_{ijt})$ is the log of the audit fee client i pays to auditor j in period t. Parameter α_2 captures the marginal willingness to pay the audit fees. Finally, ξ_{jt} denotes unobserved auditor j's attributes, e.g., reputation and quality. As in the standard discrete choice model, unobserved attributes are those that are difficult to measure or to discern in publicly available data but that clients can observe when choosing their auditor.

The discrete choice approach accounts for rich client heterogeneity. It captures the match between the audit firm and the client through interaction terms between the dummy variable for each auditor and client characteristics τ_{it}. We expand τ_{it} as follows. First, we include client's size ($Ln(Assets)$), complexity ($Ln(Segments)$), and geographical reach ($ForeignSales$). Second, we capture client's leverage ($Debt$) and profitability (ROA). Third, we consider client's working capital ($Inventory$+$Receivables$ and $Payables$). Fourth, we consider whether the client has hired the auditor in the prior year ($PriorClient$) and for how long ($Ln(YearsClient)$) and whether the auditor has an office in the same Metropolitan Statistical Area (MSA) as the client ($OfficeinMSA$). We include client's industry using the Fama-French 10-industry classification capturing, for example, that a specific auditor might be less preferred in a specific industry due to concerns of information spillover effects (e.g., Aobdia, 2015). In addition to the variables used by Gerakos and Syverson (2015). we include client's financial distress ($Loss$) and growth opportunities (BTM).

We decompose the key variable industry specialization into two levels, the firm level (ISP_firm) and the office level (ISP_office). We define an auditor to be a specialist at the firm level if the annual market share within a specific SIC2 industry is more than 30%

[7] Please note that we use several industry classifications in our study to ensure that the broadness of the industry classification matches the purpose. Following prior literature on auditor industry specialization, we use SIC2-industry classifications as a relatively fine measure for identifying industry specialists. Following Gerakos and Syverson (2015), we use the broader Fama-French 10-industry classification system to allow for differences in preferences for audit firms across industries in the demand estimation. To ensure that M&A activity matters for audit fees, we use the finer SIC3-industry classification for calculating the instrumental variable in the audit fee model.

[8] We include dummies for each Big 4 audit firm as fixed effects because the large number of interactions of these dummies with client characteristics provide sufficient flexibility for brand effects to vary across clients. Gerakos and Syverson (2015) include dummies for each Big 4 audit firm as random effects to allow brand effects to vary across clients. Our results are robust when applying the same approach.

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in terms of audit fees. Similarly, we define an auditor to be a specialist at the office level if the annual market share within a specific SIC2 industry is more than 30% in terms of audit fees at the MSA level. As an additional auditor attribute, we measure audit office size based on the total volume of annual audit fees of an audit firm within an MSA.

Measuring industry specialization based on market shares is commonly done in audit research but may trigger an endogeneity concern. The concern is that being an industry specialist could be mechanically related to the likelihood of being selected because industry specialists tend to have more clients.[9] Accordingly, being an industry specialist based on market shares might reflect those client characteristics that determine the demand for this specific auditor (e.g., Minutti-Meza, 2013). Our empirical approach addresses this concern by controlling flexibly for client heterogeneity. Furthermore, the estimation of conditional choice probabilities absorbs client-specific heterogeneity that does not vary over audit firm choice, comparable to an implicit fixed effect at the client level (Gerakos and Syverson, 2017).

In the baseline model, we include auditor industry specialization as a main effect to identify the first-order demand for auditor industry specialization. Later on, we add interaction effects between auditor industry specialization and client characteristics to test for the second-order effects of differential demand.[10] The overall utility of client i choosing auditor j in period t is represented by u_{ijt}:

$$u_{ijt} = V_{ijt} + \epsilon_{ijt} \tag{2}$$

Variable ϵ_{ijt} is used to capture random preference shocks during the process of hiring the audit firm. By definition, these random preference shocks are unobservable for the researcher but observable for the client.

We assume that ϵ_{ijt} is iid type 1 extreme value distributed. With this assumption, the probability Pr_{ijt} of client i choosing auditor j in period t conditional on client's and

[9] This relation is partially mitigated by using a market share approach based on audit fees instead of the number of clients.

[10] We include interaction effects between auditor brand name and client characteristics in the baseline model. The reason for including these interaction effects is that client preferences are expected to be heterogeneous for the auditor brand name because it is a general attribute not related to client characteristics. In comparison, auditor industry specialization is a less general attribute partly related to client characteristics because it is specific to the respective industry. Therefore, we set up a parsimonious and convergent model by not including interaction terms (baseline model) and only selected interaction terms based on theory (differential demand model) between auditor industry specialization and client characteristics. Additional tests using a parsimonious model without any interaction effects show that the relative size of the value of auditor industry specialization and auditor brand name is not driven by the different number of interaction terms included in the baseline and the differential demand model.

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auditor's attributes is as follows (McFadden, 1974):

$$Pr_{ijt} = \frac{exp(V_{ijt})}{\sum_{k=0}^{4} exp(V_{ikt})}. \tag{3}$$

As the functional form of equation (3) indicates, the probability of choosing a specific audit firm increases monotonically with the utility associated with the choice. Given the form of the choice probabilities, we can estimate the parameters of the utility function by maximizing log likelihoods (see Gerakos and Syverson, 2015, for more details).

3.2 Audit fees

Our demand estimation requires audit fee data for all possible client-auditor pairs. However, it is only possible to observe audit fees paid by the client for the audit firm selected (realized choice) but not for the audit firm not selected (unrealized choice). Fortunately, it is possible to estimate audit fees with large precision using client characteristics as determinants in a standard audit fee model. Following Gerakos and Syverson (2015) and Guo et al. (2017), we use the parameters derived in the estimation of the standard audit fee model to predict audit fees for all client-auditor pairs, including both realized and unrealized choices. This approach implicitly assumes that the rule of setting audit fees is common knowledge for clients and auditors. Using predicted audit fees instead of observed audit fees for realized choices avoids the risk of inducing a negative price shock that would bias our estimation (Gerakos and Syverson, 2015). One remaining risk is that selection issues might induce systematic differences in audit fees between realized and unrealized choices. As our estimates are only based on the audit fees of realized choices, our estimates for unrealized choices might be biased. Controlling for this issue would require information about bidding prices submitted in the auditor selection process which is not available to us. Nevertheless, as long as audit fees are largely a function of client characteristics, the issue of missing fees should be less severe than in other settings, e.g., missing prices in the setting of household purchases (e.g., Erdem et al., 1998).

One important issue is the endogeneity of audit fees for audit demand (equation (1)). Audit fees are endogenous if they are correlated with auditor attributes that are not perfectly observable for the audit researcher, e.g., audit quality or auditor reputation ($cov(\ln(p_{ijt}), \xi_{jt}) \neq 0$). An instrumental variable method can resolve endogeneity issues. Following Gerakos and Syverson (2015) and Guo et al. (2017), we use the supply shock

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caused by client firm mergers and acquisitions (M&A) within an industry in the prior year as an instrument z_{it-1} (*MA_SIC3*). This instrument should be relevant in our context because one of the firms involved in the merger has to drop its auditor. The auditor losing the client will face excess capacity and will start competing more fiercely, inducing a downward pressure on fees (e.g., Christiansen and Loft, 1992; Francis et al., 2017). Based on this connection of our instrument to the supply side, we consider it to be exogenous to audit demand.[11]

We use a control function approach to correct for endogeneity (Petrin and Train, 2010; Wooldridge, 2015). The idea of this approach is to derive a variable that, when included in the empirical model, renders the variation of the endogenous variable exogenous (Heckman and Robb, 1985). The approach follows a two-stage procedure. In the first stage, the endogenous variable is regressed on a set of control variables including an instrument. In the second stage, the residual of the first-stage regression is included in addition to the predicted value of the endogenous variable. Reflecting this procedure, the control function approach is also called a two-stage residual-inclusion method (2SRI). It is a modified version of the two-stage least squares (2SLS) method used in linear models and the two-stage predictor-substitution model (2SPS) used in non-linear models with instrumental variables. We use a control function approach because it provides consistent estimates for discrete choice models, whereas the standard two-stage predictor substitution approach does not (e.g., Amemiya, 1974; Terza et al., 2008).

An alternative approach to the control function method is the product-market controls method introduced by Berry et al. (1995). This approach is widely used to elicit demand and cost parameters using market-level data. The idea of this approach is to compare observed market shares to market shares predicted by the model to identify unobserved demand factors. The control function approach is suitable if micro-level data on client purchases is available because it reduces computational complexity and provides more robust estimates than the product-market controls method (e.g., Petrin and Train, 2010). Accordingly, the control function is more suitable for our setting.

Applied to our setting, the control function approach requires us to derive a proxy for the

[11] We address the concern that companies' M&A activities might be associated with auditor demand by performing a robustness test that excludes companies involved in M&A activities in the year where these activities take place and in the following year. We find that our instrument remains relevant and that the results of the audit fee model remain robust (see Online Appendix Table A5 and Table A6). In the main tests, Gerakos and Syverson (2015) use the supply shock from the demise of Arthur Andersen in 2002 as an instrument. As the relevance of the instrument diminishes over time, we refrain from using it for our study that includes a later period.

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unobserved variable ξ_{jt} that, when included in the demand estimation in the second stage, renders the variation of the endogenous variable $\ln(p_{ijt})$ exogenous. We derive this proxy from the residual of the first stage model where the endogenous variable $\ln(p_{ijt})$ is regressed on a set of control variables including an instrument z_{it-1}. We use the ratio of three-digit SIC industry assets merged in the prior year (*MA_SIC3*) to instrument the variation of audit fees in the current period. The empirical model of the first stage regression is as follows:

$$\ln(p_{ijt}) = \alpha_0 + \alpha_1 z_{it-1} + \alpha_2 ISP_{ijt} + \sum_r \rho_r \tau_{rit} + \Gamma_t + \Gamma_j + \Gamma_{ind} + \mu_{ijt} \tag{4}$$

As the discrete choice model is essentially a non-linear model, our application of the control function approach requires the conditional independence assumption (Blundell and Powell, 2003):[12]

$$E[\xi_{jt}|z_{it-1}] = 0 \text{ and } E[\mu_{ijt}|z_{it-1}] = 0$$

We implement the audit fee model using a standard OLS regression with a linear function of client's characteristics ($\sum_r \rho_r \tau_{rit}$) and the iid error term μ_{ijt}.[13] We use largely the same set of control variables as Gerakos and Syverson (2015) with four modifications. First, we add two measures commonly used in audit fee studies (Hay et al., 2006), namely client's financial distress (*Loss*) and liquidity ($CurrentRatio$). Second, we do not include a variable capturing whether the firm was a client of the audit firm in the preceding period given recent research questioning the existence of fee discounting in initial year audit engagements (Barua et al., 2020).[14] Third, we include our measures for auditor industry specialization (ISP_firm and ISP_office). Therefore, any audit fee premiums associated with industry specializations are captured in our audit fee model. Fourth, we include fixed effects for fiscal year (Γ_t), audit firm (Γ_j), and industry (Γ_{ind}).

Table (3) presents the results of the audit fee model using basically the same sample as for demand estimation (see section 4.1).[15] As expected, the coefficient for the instrument is negative and highly significant, suggesting that M&A activities within the industry lead to supply-induced price pressure in the following year. Moreover, the results show that

[12] This assumption is a stronger exogeneity condition than the unconditional moment restriction ($E[z'_{it-1}\xi_{jt}] = 0$ and $E[z'_{it-1}\mu_{ijt}] = 0$) required for the linear model (Blundell and Powell, 2003).

[13] We estimate the audit fee model with the correction procedure as in Wooldridge (2003).

[14] Our results remain robust when including the same set of control variables in the audit fee model as Gerakos and Syverson (2015) (Online Appendix Table A5 and Table A6).

[15] The only difference is that we do not expand the sample to include unrealized choices but use only realized choices in the audit fee model.

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the included auditor attributes on auditor industry specialization and audit office size are associated with higher audit fees. The direction of the effect of the control variables is consistent with prior literature. Figure (1) shows that the predicted audit fees fit the actual audit fees well. The correlation between predicted audit fees and actual audit fees ranges from 0.833 to 0.886 for each of the audit firms in the choice set.

We use parametric OLS regressions to estimate audit fees (equation (4)). An alternative approach would be to use nonparametric methods, e.g., random forests. Nonparametric methods have the potential to achieve a better fit. However, the asymptotic properties of these methods are not well understood. For example, there is some initial evidence that random forests can generate consistent results, but it is restricted to the specific case of additive regression models (Scornet et al., 2015). Therefore, we refrain from using them to avoid the risk of inconsistent estimates.[16]

After the first stage regression of equation (4), residual $\hat{\mu}_{ijt}$ enters the demand estimation as a proxy variable in the second stage. The control function approach allows for a flexible functional form of the variable $\hat{\mu}_{ijt}$:

$$\xi_{jt} = h(\hat{\mu}_{ijt})$$

where $h(\hat{\mu}_{ijt})$ denotes the control function.

We use a linear functional form for $h(\hat{\mu}_{ijt})$ as it is commonly done in the literature (Wooldridge, 2015).[17] We define ξ_{jt} as the arithmetic average across all clients for the same audit firm j in our demand estimation.[18] Our specification assumes that the unobserved variable ξ_{jt} does not vary across clients of the same audit firm. From a conceptual perspective, this assumption is reasonable when the unobserved variable represents auditor attributes, e.g., auditor reputation. From an empirical perspective, this assumption is necessary because it is not possible to derive a proxy for ξ for each auditor-client-year combination from our audit fee model. The reason is that the audit fee model only in-

[16] In a robustness test, we apply random forests with fivefold cross-validation (see Online Appendix Figure A1). We observe that the estimation results at the second stage remain largely robust with the main difference being a smaller coefficient for the audit fee variable (see Online Appendix Table A7). The coefficients for our main variables of interest are basically unchanged: The coefficients when using random forests for predicting audit fees are 0.6462 for ISP_firm and 1.5289 for ISP_office, compared to 0.6873 for ISP_firm and 1.5869 for ISP_office when using OLS. Using random forests, the correlation between predicted audit fees and actual audit fees ranges from 0.888 to 0.947 for each of the audit firms in the choice set.

[17] In a robustness test, we include the residual $\hat{\mu}_{ijt}$ additionally in a quadratic functional form and find that the results remain robust (see Online Appendix Table A8).

[18] We perform an additional test where we weight the residual $\hat{\mu}_{ijt}$ by client size in terms of total assets and find that the results remain robust (see Online Appendix Table A8).

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cludes audit fees from realized but not from unrealized auditor choices. However, we still allow for rich client heterogeneity in our demand estimation by including a large number of interaction terms with client characteristics and auditor attributes.[19]

4 Demand estimation

4.1 Sample and descriptive statistics

Our sample consists of US-listed firms from 2003 to 2018 (see Table (1) for a description of the sample selection). We use Audit Analytics as our data source for audit fees and auditor-client matches and include all data available for our sample period as the starting point. Following prior research, we exclude financial institutions. The data on client attributes are from Compustat. We exclude observations with missing data in Compustat, non-US firms, and shell firms. Further, we exclude observations with missing values of the MSA of the auditors' office and with other missing values needed for the audit fee model and the demand estimation. The final sample includes 49,962 observations. We obtain data on mergers and acquisitions from the Thomson One Banker database.

Table (2) presents the descriptive statistics.[20] There are three groups of variables: (1) auditor attributes, e.g., industry specialist at office level (*ISP_office*), industry specialist at firm level (*ISP_firm*); (2) auditor-client attributes, e.g., audit fee (*Ln(Audit Fees)*), auditor tenure (*Ln(Years Client)*), and (3) client attributes, e.g., client size (*Ln(Assets)*), number of segments (*Ln(Segments)*). All variables are defined in the Appendix.

The final sample includes only realized choices. We use this sample to estimate the audit fee model as described above. For performing the demand estimation, we expand the sample by creating additional observations by pairing each client-year observation with all unrealized choices of auditors within the choice set. As we have five auditors in the choice set (each of the Big 4 and the group of non-Big 4 auditors), the sample size for demand estimation is five times the final sample.[21]

[19] Petrin and Train (2010) discuss less restrictive specifications of the error structure. We would like to point out that these specifications are often used to account for individual heterogeneous preferences in cases where only market-level data is available. As we have detailed micro-level data about client characteristics available that we include in our demand estimation, we believe that our model already captures a rich form of individual heterogeneity. Therefore, we do not consider it necessary to add another level of unobserved heterogeneity in the error term. Nevertheless, we perform a robustness test specifying the error structure as in example 2 of Petrin and Train (2010) to be $\lambda\mu_{ijt} + \sigma\eta_{ijt} + \epsilon_{ijt}$. Our results remain robust (see Online Appendix Table A8).

[20] To ensure that extreme values do not interfere our results, we winsorize the data at the 1% and 99% levels.

[21] In a few specific cases (<0.1% of all cases), the choice set is smaller because we remove auditors that

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4.2 First-order demand for auditor industry specialization

Table (4) presents the results of the demand estimation. The results reveal a significant first-order demand for industry specialization at both the audit firm (ISP_firm) and the audit office level (ISP_office), implying that clients, on average, prefer hiring an industry specialist. This finding is present when including industry specialization at the firm and office level both separately (Model I & II) and jointly (Model III).

The observed demand for industry specialization is incremental to the demand for a specific audit firm. It is possible to identify this incremental effect because we include auditor fixed effects into the model to control for auditor brand name.[22] Furthermore, we control for audit office characteristics by including audit office size as an attribute (AO_size). We observe that clients prefer hiring an auditor with a larger office.

The results for the auditor attributes and the auditor-client match are largely consistent with prior research (Gerakos and Syverson, 2015).[23] We observe a negative coefficient for audit fees (*Ln(Audit Fees)*) and a positive coefficient for the incumbent auditor (*Prior Client*). The residual variable (*Residual*) from the first stage of our control function approach is insignificant. Furthermore, the model shows that clients have heterogeneous demands for auditors along with a number of client characteristics. Consistent with Gerakos and Syverson (2015), we find that clients with a larger proportion for foreign sales (*Foreign Sales*), lower profitability (ROA), and lower levels of current assets (*Inventory + Receivables*) tend to have a stronger demand for the Big 4 audit firms. In addition, we observe that clients with a loss (*Loss*) and clients with higher growth potential (BTM) have a stronger demand for the Big 4 audit firms.

4.3 Interdependencies between auditor attributes

Next, we investigate whether auditor demand reflects interdependencies between auditor attributes. A substitutive (complementary) interdependency means that the marginal benefit of each attribute decreases (increases) as the level of the other attribute increases. The implication for audit firms is that they should focus their investments on one key auditor

resigned from auditing the client in the prior year (Gerakos and Syverson, 2015).

[22] Our demand estimation shows a negative sign for each of the Big 4 audit firms. This result needs to be considered jointly with the interaction effects of auditor fixed effects with client attributes, indicating differential demand for the Big 4 audit firms. In particular, we observe that larger clients (*Ln(Assets)*) have a stronger demand for the Big 4 audit firms. Accordingly, the negative sign for each of the Big 4 audit firms only indicates that the smallest clients prefer non-Big 4 audit firms but that larger clients prefer the Big 4 audit firms.

[23] We base the discussion of the control variables on Model III.

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attribute for substitutive relations and multiple attributes for complementary relations.

Marketing research uses the economic principle of satiation to derive expectations for the interdependency of attributes (e.g., Voss et al., 2010). Strong satiation goods include essential products where consumers' needs are quickly satiated with almost zero marginal utility from additional consumption. As consumers are already satiated when these kinds of products have some key characteristic, improvements along other characteristics do not provide additional utility, resulting in a ceiling effect and substitutive relations. Weak satiation goods include luxury goods that continue to have a positive marginal utility even when consumed at very high levels. The lack of a strict ceiling effect for these goods provides opportunities for increasing consumers' utility by improving all relevant attributes of the good, suggesting complementary interdependencies. It is an empirical question whether auditing is a weak or strong satiation good. While the mandatory nature of auditing suggests that it could be a strong satiation good, the demand of clients for audit quality beyond minimum requirements suggests that it could be a weak satiation good.[24]

We test for interdependencies by including interaction terms between auditor attributes in the demand estimation model. Table (5) presents the results. First, we find a significantly negative interaction effect between industry specialization at the firm level (ISP_firm) and the office level (ISP_office). This finding implies a substitutive relationship. That is, clients have lesser demand for industry specialization at the firm level if the auditor is an industry specialist at the office level, and vice versa. Considering the coefficients for the main effects and interaction effects, one can conjecture that the incremental demand for industry specialists at the firm level is weaker if the auditor is already an industry specialist at the office level. Second, we observe a significantly negative effect when interacting audit office size (AO_size) with either industry specialization at the firm level (ISP_firm, column I) or the office level (ISP_office, column II). These findings imply substitutive relationships. That is, clients prefer larger office sizes to a lesser degree when the auditor is an industry specialist and vice versa. Finally, we observe that the coefficients for the control variables remain largely unchanged compared to the baseline model.

We also evaluate the interdependency between industry specialization and auditor brand name. For that purpose, we expand the auditor choice set by adding the fifth and sixth

[24] Prior audit research has investigated joint effects of auditor attributes. In particular, research on auditor industry specialization finds a joint effect of industry specialization at the firm level and office level for audit fees (e.g., Francis et al., 2005) and audit quality (e.g., Reichelt and Wang, 2010). This finding suggests that there might be interdependencies between both levels but does not explicitly address whether these interdependencies are substitutive or complementary.

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largest audit firm of our sample, BDO and Grant Thornton. In our sample, these audit firms never achieve industry specialization at the firm level but they sometimes achieve it at the office level. We interact auditor industry specialization at the office level with a Big 4 dummy and observe a significantly negative effect (see Table (5)). This finding suggests that clients derive a lower utility from the industry specialization of the Big 4 relative to non-Big 4 audit firms. As the brand value of the Big 4 audit firms is presumably larger than for non-Big 4 audit firms, it implies a substitutive relation between industry specialization and brand name.

4.4 Second-order demand for auditor industry specialization

Connecting to prior research on auditor choice, we test for the second-order demand for auditor industry specialization (Table (6)). We focus on two main client attributes, client size (*Ln(Assets)*) and client complexity (*Ln(Segments)*). We extend the baseline model by interacting these attributes with our measures for auditor industry specialization at the firm and office levels. We observe that the coefficients for the control variables remain largely unchanged compared to the baseline model.

First, we find that larger and more complex clients have a significantly stronger demand for industry specialists at the office level (columns I & II). This finding is consistent with prior literature arguing that these types of firms have a greater demand for high-quality auditors to mitigate principal-agent conflicts. Going beyond prior research, our model does not only include the second-order differential effect but also the first-order main effect, enabling us to infer whether clients have, in general, a positive demand for auditor industry specialization. Our results reveal that all clients – except for the smallest – prefer to be audited by an industry specialist at the office level.

Second, we observe that larger and more complex clients do not have a significantly stronger preference for industry specialization at the firm level, when both firm and office level specialization are included in the model (columns I & II). This finding contradicts prior studies, which usually observe a positive relation. The reason for our novel finding is presumably that our discrete choice model application goes beyond prior research by explicitly controlling for auditor industry specialization at the audit office level and auditor brand name effects. When we do not control for the audit office level, we observe a significantly positive interaction of client size and audit firm industry specialization (untabulated). Further, we find a positive relation between larger clients and the likelihood of

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hiring an industry specialist at the firm level when employing a binary choice model based on prior research (untabulated). Therefore, our application of the discrete choice model reveals that the differential demand for auditor industry specialization is mainly driven by industry specialization at the office level and not by industry specialization at the firm level.

4.5 Robustness tests

In our main analyses, we classify auditors as industry specialists based on a market share threshold of 30%. In a robustness test, we define only the audit firm (audit office) with the largest market share at the firm (MSA) level to be an industry specialist. Table (7) shows that the results of our baseline model (column I) and the results for interdependencies (columns II & III) remain largely unchanged.[25]

Another issue is that the use of market share measures for auditor industry specialization may inherently incorporate the size of the market into the industry specialist measure. For example, if an audit market at the MSA-industry level only includes one client, the audit office in charge of that client would automatically be designated as a specialist. Therefore, we perform a robustness test excluding all markets at the MSA-industry level with only one client. Table (8) shows that the results remain robust.

Third, we examine the robustness of our results to the specification of the choice set at the audit office level. As our study is the first one integrating audit office attributes into the discrete choice model, we develop our own novel approach. In our main analyses, we include the offices in the choice set that are located in the MSA of the client's headquarters. In the special case where the client chooses an auditor outside its MSA, we include only the offices in the choice set that are located in the MSA of the chosen auditor. To test whether our results are sensitive to that approach, we re-perform our analysis excluding all clients choosing auditors outside their MSA. Table (9) shows that the results remain robust.

Finally, we test the robustness of our results to explicitly model tradeoffs involved in the first-order demand for auditor industry specialists. In our main analyses, we consider tradeoffs to some degree, e.g., by including auditor industry specialization into the audit fee model. In this robustness test, we examine whether our results are sensitive to considering the risk of information spillovers. Previous research suggests that higher market competition is associated with higher levels of information spillover risk (e.g., Aobdia, 2015; Bills et al., 2020). Therefore, we control for this risk by including the Herfindahl index computed at

[25] In untabulated analyses, we also test the robustness of our findings for second-order effects. We find that the results remain robust for all modifications discussed in this subsection.

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the SIC2 level in our demand estimation and interacting the index with individual auditor fixed effects. Table (10) shows that the results remain robust.[26]

5 Counterfactual analyses

5.1 Quantifying the value of auditor industry specialists

5.1.1 Method

In this section, we estimate the value of auditor industry specialization for audit clients based on the parameters derived in the demand estimation. Our basic approach is to calculate and decompose the utility loss clients experience when being restricted from enjoying the benefits from industry specialization.

We use the approach of McFadden (1999) to derive the monetary value of auditor industry specialization. This approach identifies how much compensation is needed to make the client indifferent between two choices. Suppose that client i chooses auditor k to maximize utility u_{ikt} in reality. Further, suppose that the same client i chooses auditor j to maximize utility u_{ikt} in a counterfactual scenario. Then, it is possible to calculate the amount of money C_{ijt} that is needed to render the client indifferent between the real and the counterfactual choice. Aggregating these amounts across clients provides us with an estimate for the total value of industry specialization per year.

The equation for balancing clients' utility derived from auditor choice in the real and counterfactual world is as follows.[27]

$$\max_k u_{ikt}(p_{ikt}, \vec{x}_{ikt}, \epsilon_{ikt}) = \max_j u_{ijt}(p_{ijt} - C_{ijt}, \vec{x}_{ijt}, \epsilon_{ijt}) \tag{5}$$

where p_{ikt} is the audit fee, ϵ_{ikt} is the idiosyncratic shock, and the vector $\vec{x}_{ikt}$ represents all the other factors affecting client i's utility of choosing any auditor k in period t.

We solve for C_{ijt} that equates these two maximized utilities. Given the log form of audit fees in the utility function, C_{ijt} is the exponent of the difference in maximized utilities between the two regimes transferred by the estimated marginal willingness to pay for each individual client.

[26] In untabulated analyses, we interact the Herfindahl index additionally with our measures for industry specialization and find that the results remain robust.

[27] Note that we use the predicted auditor choice both in the real and the counterfactual world to have a consistent benchmark, disentangling effects from auditor industry specialization from prediction errors. However, since the prediction errors are small, the results do not change substantially when using the actual auditor choice for the real-world setting.

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We simulate clients' auditor choice in the counterfactual scenario by assuming that clients choose the auditor associated with the highest overall utility (see equation 2). We obtain the deterministic part of the utility based on the estimated parameters and the observable client and auditor characteristics from our baseline model. We randomly draw the unobserved part of the utility ϵ_{ikt} from an iid extreme value distribution. We repeat the random draw of this idiosyncratic preference shock 500 times for each client-auditor-year combination.[28] For each draw, we calculate the resulting overall utility. Based on the information on auditor choice in the actual and the counterfactual scenario, we calculate the value of C_{ijt} using equation (5). We obtain the final value of C_{ijt} by averaging each calculated value across the 500 estimations. We aggregate the average estimates across all clients to obtain the value of industry specialization.

5.1.2 Estimation

We quantify the value of industry specialization in a counterfactual scenario that restricts clients from hiring an industry specialist. This approach follows prior studies in industrial organization that estimate the value of specific attributes of products by removing options from the choice set (e.g., Petrin, 2002). One advantage is that we can use the same general approach to calculate the utility loss resulting from other auditor attributes as a benchmark, e.g., auditor brand name. Furthermore, this counterfactual scenario is realistic as it shares similarities to an external rotation requirement in the case that the incumbent auditor is an industry specialist.

We implement this counterfactual setting by considering the specific level of industry specialization. We restrict clients originally hiring an office-level (firm-level) industry specialists from hiring an office-level (firm-level) industry specialist again.[29] The utility loss clients experience in this counterfactual does not only results from industry specialization but also from other auditor attributes. Therefore, we decompose the utility difference C_{ijt}

[28] As Train (2009, p. 217) describes "The use of independent draws in simulation is appealing because it is conceptually straightforward and the statistical properties of the resulting simulation are easy to derive." The goal of multiple draws is to make sure that the average realization converges to zero. Consistently, a robustness test where we set the preference shock ϵ_{ikt} to zero (instead of randomly drawing it) yields almost identical results in our setting (see Online Appendix Figure A3).

[29] For this counterfactual scenario, we observe that clients originally hiring an industry specialist at the office level switch to an auditor with industry specialization at the firm level in 5.85% and to a non-industry specialist in the remaining 94.15% of the cases. Furthermore, clients originally hiring an industry specialist at the firm level switch to an auditor with industry specialization at the office level in 32.60% and to a non-industry specialist in the remaining 67.40% of the cases. In an alternative scenario, we restrict clients originally hiring an industry specialist from hiring any type of industry specialist again. We find that our estimates for the value of auditor industry specialization are only slightly larger compared to our main scenario (see Online Appendix Figure A2).

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into two parts, C^1_{ijt} and C^2_{ijt}:

$$C_{ijt} = C^1_{ijt} + C^2_{ijt}$$

The first component C^1_{ijt} represents the difference derived from the change of specific auditor attributes, namely industry specialization at the firm and office level. The second component C^2_{ijt} represents the remaining difference due to the change of all other auditor attributes, i.e., client-auditor matching.

Figure (2) shows the quantification of the value of industry specialization using this counterfactual. As Panel A shows, we find that the total value of industry specialization is 1.9 million US Dollars (0.13% of audit fees) at the firm level and 3.3 million US Dollars (0.23% of audit fees) at the office level when aggregating across all clients on a yearly basis.[30]

The value of industry specialization represents the aggregate per-year values of C^1_{ijt}. The aggregate per-year values of C^2_{ijt}, capturing the value of all other auditor attributes, are substantially larger, i.e., 211.1 (379.5 million US Dollars) at the audit firm (audit office) level.[31] When interpreting these results, it is important to consider that the selection of an auditor is a joint choice of industry specialization and other auditor attributes. Therefore, restricting clients from hiring an industry specialist has a large impact on clients even though the direct effect of industry specialization is small.

As another benchmark, we derive the value of the auditor brand. We obtain it for each Big 4 audit firm by implementing a counterfactual setting that excludes each Big 4 audit firm at a time from the choice set. We use the same approach as described above to quantify values. We find that the value of auditor brand name is substantially larger than the value of industry specialization, ranging from 232.3 million US Dollars to 590.5 million US Dollars (16.4% to 41.68% of audit fees). This value includes both the benefit from hiring the specific audit firm (e.g., brand loyalty) and the foregone need to switch the auditor (e.g., switching costs).

Our counterfactual approach of restricting clients from hiring their incumbent auditor if it is an industry specialist shares similarities to a mandatory rotation requirement. Gerakos and Syverson (2015) investigate this scenario and find a substantial reduction in consumer surplus of 24.5% of audit fees in their baseline scenario. This percentage value is comparable

[30] The value accrues to the clients with an industry specialist as an auditor. Scaling the value of industry specialization only by the audit fees paid by clients hiring an industry specialist, we find that the value of industry specialization is 0.39% at the audit firm level and 0.35% at the audit office level.

[31] The results are illustrated in Online Appendix Figure A4.

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to our estimates for auditor brand value. Going beyond Gerakos and Syverson (2015), our study shows that clients having to switch auditors experience losses in consumer surplus largely because they have to switch away from their most preferred auditor brand name. In comparison, the loss in consumer surplus clients suffers because they have to switch away from an auditor industry specialist is relatively small.

Additional analyses show that the value of industry specialization differs across auditor and client types. Figure (2), Panel A, illustrates that it ranges from 0.2 to 0.8 million US Dollars at the firm level and from 0.5 to 1.1 million US Dollars at the office level when aggregating across all clients of each audit firm separately. Panel B reveals that the aggregate value of industry specialization is higher for larger clients (1.5 (2.5) million US Dollars at the firm (office level)) than for smaller clients (0.4 (0.7) million US Dollars at the firm (office level)) when performing a median split based on clients' assets. Finally, Panel C shows that the value of industry specialization differs across industries based on the Fama-French 10-industry classification.[32]

5.1.3 Robustness tests

In a first robustness test, we quantify the value of industry specialization using an alternative counterfactual scenario. We implement it by setting the coefficients for industry specialization at the firm and office level to zero. Consistently, we also set the fee premium charged by industry specialists to zero. This specification reflects a setting where the value of industry specialization vanishes or diminishes, e.g., through lower investments by audit firms in response to regulatory changes (e.g., Petrov and Stocken, 2019). We allow clients to switch auditors in response to this change and find that 4.34% (6.03%) of clients originally hiring an industry specialist at the firm (office) level switch.

We derive the value of industry specialization by comparing the utility obtained from the original auditor to the utility obtained from the new auditor in the counterfactual scenario. We obtain a value of industry specialization of 1.9 million US Dollars (0.13% of audit fees) at the firm level and of 3.3 million US Dollars (0.23% of audit fees) at the office level. These results are very similar to the results of the first counterfactual, suggesting that our results are robust.

[32] In our counterfactual analyses, we allow the value for industry specialization to differ across clients. However, note that we focus on the first-order demand for industry specialization. Therefore, we perform the counterfactual analysis based on the estimates from our baseline model that includes industry specialization as a main effect but not as an interaction effect. Accordingly, the main source of variation is whether clients hire an industry specialist in the real-world setting.

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In a second robustness test, we use the same counterfactual scenario as in our main test but base it on a homogeneous demand estimation model.[33] The main test is based on the baseline demand estimation allowing for heterogeneous preferences for specific auditor brand names contingent on client characteristics. Meanwhile, model limitations restrict us from integrating similarly heterogeneous preferences for auditor industry specialization. Therefore, one concern could be that these differences in handling auditor brand names and auditor industry specialization influence the relative value of these auditor attributes. We address this concern by setting up a homogeneous demand estimation model that includes auditor attributes solely as main effects but not in interaction with client characteristics. That is, we assume that only auditor attributes determine auditor choice and that the combinations of auditor attributes and client characteristics do not matter. In other words, this specification imposes the assumption that all clients have a homogenous preference over auditor brand name. As this assumption is not consistent with the choice pattern we observe in the data, the results of this robustness test should be interpreted with caution.

Using the homogenous demand estimation, we continue to find that the value of industry specialization is substantially smaller than the value of brand name. We observe a value of 12.7 million US Dollars (24.5 million US Dollars) for industry specialization at the audit firm (audit office level), compared to brand values ranging from 707.2 million US Dollars to 1163.9 million US Dollars. These values are larger compared to the values obtained in our main model because the coefficient for audit fees is smaller than in the baseline model implying a lower sensitivity of clients to audit fees. Nevertheless, we can conclude that our finding that the value of industry specialization is relatively small compared to auditor brand value is robust.

5.2 Estimating the effects on audit market share

We examine the importance of auditor industry specialization for audit firms by investigating market share effects. For each audit firm, we set up a counterfactual analysis that assumes that it would become a specialist in all industries, either at the firm or office level. To be consistent, we predict audit fees for the respective audit firm assuming that it would earn an additional fee premium due to the change of specialist status.[34] Next, we predict

[33] Please find the estimation results in Online Appendix Table A9.

[34] We also perform the counterfactual analyses for the group of small audit firms by assuming that they constitute one audit firm. That is, we assume that all small audit firms become auditor industry specialists and we compute the change in their aggregate market share. For the small audit firms, we use the average fee premium of industry specialists at the firm level according to our audit fee model.

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clients' optimal auditor firm choice by comparing the respective utilities. Finally, we compute the market share of each audit firm based on clients' choices in each scenario. We calculate the percentage change by comparing the mean value of predicted market shares in the counterfactual world to that value in the real world.[35]

Figure (3) shows the market share changes in terms of the number of clients. In Panel A, we observe that all audit firms would gain market shares when becoming industry specialists despite charging clients a fee premium for it. The increase for each Big 4 audit firm is between 3.7% and 9.4% for industry specialization at the office level and between 2.0% and 5.5% for industry specialization at the firm level. The gains for the smaller audit firms are 10.0% and 3.7%, respectively.[36] In a separate analysis, we conduct the same counterfactual analyses but without allowing specialists to charge a fee premium. As Panel B shows, we observe that the increase in market share of each Big 4 audit firm would become slightly larger, ranging from 5% to 10.8% at the office level and ranging from 2.5% to 5.5% at the firm level.

Figure (4) shows the changes in market share in terms of clients' assets. In Panel A, we observe a similar pattern for the Big 4 audit firms as above when calculating market share changes in terms of the number of clients. The increase in market shares ranges from 1.5% to 5.3% for industry specialization at the office level and from 0.5% to 3.0% for industry specialization at the firm level. Given the lower market share base for small audit firms in terms of clients' assets, their relative increases in market shares are more dramatic. It is 43% for industry specialization at the office level and 16% for industry specialization at the firm level. In Panel B, we observe a similar pattern when audit fee premiums for industry specialization are set to zero. The market share increase of each Big 4 audit firm would be slightly larger, ranging from 2.1% to 6.0% at the office level and from 1.0% to 3.0% at the firm level. It indicates that the impact of industry specialization on market structure remains similar whether or not the fee premium is considered.

[35] We use the predicted market shares for the real-world setting to have a consistent benchmark, disentangling effects from auditor industry specialization from prediction errors. However, since the prediction errors are small, the results do not change substantially when using actual market shares instead.

[36] We use this counterfactual to calculate the gain in consumer surplus for clients. As indicated in Online Appendix Figure A5, the per-year value aggregated across all auditors is 0.6 million US dollars (0.04% of audit fees) at the audit firm level and 1.0 million US dollars (0.07% of audit fees) at the audit office level. These values accrue to clients not hiring an industry specialist originally.

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6 Conclusion

This study uses demand estimation techniques to investigate the role of auditor industry specialization. The application of the discrete choice model expands the opportunities for investigating the demand for industry specialization significantly compared to the traditionally used binary choice model. Furthermore, it is possible to set up a structural model based on the demand estimates to perform counterfactual analysis for assessing the economic importance of auditor industry specialization.

Our results provide novel evidence on the role of auditor industry specialization for auditor demand. Using a discrete choice model, we find that industry specialization at both the firm and the office level has a positive first-order effect. We observe that clients perceive industry specialization at the firm and the office level as substitutes. We also provide evidence for a differential demand for industry specialization with larger and more complex clients having a stronger preference for industry specialization at the office level.

In counterfactual analyses, we assess the economic implications of the demand for auditor industry specialization. We find that the overall consumer surplus loss due to restricting clients from enjoying the benefits of industry specialization or removing industry specialists from the choice set is 5.2 million US Dollars per year (0.36% of audit fees). This number is relatively small compared to our estimates for the value of the auditor brand name. In addition, we find that industry specialization has a limited effect on the audit market structure. Audit firms developing industry expertise in all industries would gain market shares in terms of the number of clients, ranging from 3.7% to 10% for industry specialization at the office level, and ranging from 2.0% to 5.5% for industry specialization at the firm level.

One limitation of our study is that we consider a static demand estimation framework. We do not include dynamic features of the audit market that might affect auditor choices, for instance, strategic pricing or learning by audit firms. To alleviate concerns, we integrate some features into our model covering aspects of the dynamic market. For example, we consider how the pricing of industry specialization affects auditor choice by quantifying the effect of the associated audit fee premium. Further, we include measures of auditor tenure in our demand estimation model to capture the dynamic character of auditor choice.

Finally, it is important to bear in mind that our quantification of the value of industry specialization focuses on the clients perspective. Clients can obtain value from the service quality provided by the industry specialist or from the possibility of signaling a commit-

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ment to reliable financial reporting. Our model does not capture the value of industry specialization that accrues to other parties. That is, it does not include the benefits that investors enjoy from making better-informed decisions due to more reliable reporting. Further, it does not take into account the supply-side benefits of industry specialization, e.g., economies of scale.

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Tables

Table 1: Sample construction for the US audit market

Audit Analytics 2003-2018	138,341
-Exclude financial institutions	31,483
-Exclude observations missing in Compustat	38,410
Sample after merging with Computstat	68,448
-Exclude non-US firms and shell firms	11,223
-Exclude observations with missing value for auditors' MSA	5,595
-Exclude observations with missing values for relevant variables	1,668
Final sample	49,962

Notes: This table illustrates how our sample for the US audit market is constructed.

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Table 2: Descriptive statistics

	Auditor attributes				
Variable	**Mean**	**SD**	**P10**	**P90**	**N**
ISP_firm	0.2087	0.4064	0	1	49,962
ISP_office	0.4358	0.4959	0	1	49,962
AO_size	9.2184	2.0134	6.5665	11.6952	49,962
PwC	0.1640	0.3703	0	1	49,962
EY	0.1996	0.3997	0	1	49,962
Deloitte	0.1413	0.3483	0	1	49,962
KPMG	0.1318	0.3383	0	1	49,962
	Client attributes				
MA_SIC3	0.0365	0.7142	0	0.0665	49,962
Ln(Assets)	12.3123	2.8358	8.7146	15.6692	49,962
Ln(Segments)	0.9822	0.3775	0.6931	1.6094	49,962
Foreign Sales	0.1791	0.2618	0	0.5953	49,962
Debt	0.4085	1.0929	0	0.6364	49,962
ROA	-0.5104	2.2823	-0.7869	0.1193	49,962
Loss	0.4289	0.4949	0	1	49,962
Inventory + Receivables	0.2365	0.1941	0.0168	0.5171	49,962
Payables	0.1594	0.4639	0.0133	0.2339	49,962
Current Ratio	2.8415	3.2115	0.5882	5.8244	49,962
BTM	0.2910	1.3496	-0.0685	1.0574	49,962
	Auditor-client attributes				
Ln(Audit Fees)	6.4346	1.5475	4.3371	8.3455	49,962
Ln(Years Client)	1.4601	0.8388	0	2.5649	49,962
Prior Client	0.9609	0.1940	1	1	49,962
Office in MSA	0.0720	0.2585	0	0	49,962

Notes: This table provides descriptive statistics for the variables used in our main analyses. All continuous variables are winsorized at the 1% and 99% levels. All variables are defined in the Appendix.

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Table 3: Audit fee model

Ln(Audit Fees)	
MA_SIC3	-0.0101***
	(0.0016)
ISP_firm	0.0252*
	(0.0136)
ISP_office	0.0558***
	(0.0124)
Ln(Assets)	0.4602***
	(0.0039)
Ln(Segments)	0.2544***
	(0.0172)
Foreign Sales	0.5288***
	(0.0256)
Debt	0.0504***
	(0.0067)
ROA	-0.0612***
	(0.0036)
Loss	0.1823***
	(0.0099)
Inventory + Receivables	0.2791***
	(0.0356)
Current Ratio	-0.0246***
	(0.0017)
Ln(Years Client)	-0.0129**
	(0.0061)
Office in MSA	-0.1982***
	(0.0220)
Constant	0.2027***
	(0.0488)
Auditor fixed effect	Yes
Year effect	Yes
Industry fixed effect	Yes
Observations	49,962
Adjusted R^2	0.8832

Notes: This table presents the results of an OLS regression of audit fees on M&A activity in the industry (SIC3) and other control variables. The dependent variable is the natural logarithm of audit fees. The M&A activity is measured by the weighted proportion of firms involved in M&A activity in terms of assets in each industry in the preceding year. The other independent variables are measured in the same year as the audit fees.

All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 4: Demand estimation: Baseline

Model	I	II	III
ISP_firm	1.0475***		0.6873***
	(0.0590)		(0.0623)
ISP_office		1.6653***	1.5869***
		(0.0462)	(0.0471)
AO_size	0.4164***	0.3251***	0.3194***
	(0.0197)	(0.0185)	(0.0185)
Ln(Audit Fees)	-0.4497**	-2.3506***	-2.6673***
	(0.1897)	(0.2152)	(0.2244)
Residual	-5.4565***	-1.5747	-1.1848
	(1.0095)	(1.0276)	(1.0320)
Prior Client	2.9962***	3.0797***	3.0852***
	(0.0651)	(0.0636)	(0.0640)
PwC	-16.6774***	-17.3274***	-17.6886***
	(0.5640)	(0.5807)	(0.5907)
EY	-14.9286***	-14.8734***	-14.9121***
	(0.4905)	(0.5018)	(0.5040)
Deloitte	-14.4278***	-15.8484***	-16.1000***
	(0.5230)	(0.5409)	(0.5473)
KPMG	-14.4126***	-14.8668***	-15.1306***
	(0.5526)	(0.6001)	(0.6041)
PwC*Ln(Assets)	0.9423***	1.0970***	1.1267***
	(0.0424)	(0.0454)	(0.0463)
EY*Ln(Assets)	0.8932***	0.9787***	0.9944***
	(0.0359)	(0.0375)	(0.0380)
Deloitte*Ln(Assets)	0.8598***	1.0483***	1.0814***
	(0.0402)	(0.0432)	(0.0439)
KPMG*Ln(Assets)	0.8635***	0.9455***	0.9767***
	(0.0398)	(0.0441)	(0.0446)
PwC*Ln(Segments)	-0.4568***	-0.4275**	-0.4124**
	(0.1688)	(0.1719)	(0.1739)
EY*Ln(Segments)	-0.5199***	-0.4719***	-0.4831***
	(0.1473)	(0.1510)	(0.1511)
Deloitte*Ln(Segments)	-0.3480**	-0.4464***	-0.4884***
	(0.1426)	(0.1452)	(0.1461)
KPMG*Ln(Segments)	-0.5666***	-0.3554**	-0.3516**
	(0.1448)	(0.1496)	(0.1492)
PwC*Foreign Sales	-0.0396	0.7575***	0.7633***
	(0.2179)	(0.2229)	(0.2279)

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Table 4 (continued)

EY*Foreign Sales	-0.3020	0.8278***	1.0123***
	(0.2049)	(0.2142)	(0.2196)
Deloitte*Foreign Sales	0.1757	1.4150***	1.6529***
	(0.2186)	(0.2240)	(0.2283)
KPMG*Foreign Sales	-0.0866	1.3119***	1.4633***
	(0.2222)	(0.2307)	(0.2333)
PwC*Debt	0.0501	0.1936	0.2251
	(0.1653)	(0.1653)	(0.1680)
EY*Debt	-0.4349***	-0.3278***	-0.3549***
	(0.1287)	(0.1188)	(0.1226)
Deloitte*Debt	-0.2165*	-0.3067**	-0.3113**
	(0.1233)	(0.1282)	(0.1288)
KPMG*Debt	0.1297	0.3997***	0.4429***
	(0.1232)	(0.1511)	(0.1514)
PwC*ROA	-0.1326	-0.3955***	-0.4207***
	(0.1268)	(0.1175)	(0.1270)
EY*ROA	-0.1486***	-0.3518***	-0.3732***
	(0.0549)	(0.0567)	(0.0578)
Deloitte*ROA	-0.1546**	-0.4503***	-0.4912***
	(0.0779)	(0.0607)	(0.0615)
KPMG*ROA	-0.1871***	-0.2210***	-0.2344***
	(0.0405)	(0.0399)	(0.0398)
PwC*Inventory + Receivables	-1.0974***	0.1220	0.2702
	(0.3321)	(0.3375)	(0.3478)
EY*Inventory + Receivables	-1.4748***	-0.0094	-0.1165
	(0.2928)	(0.3007)	(0.3048)
Deloitte*Inventory + Receivables	-0.6807**	0.1120	0.2605
	(0.2801)	(0.2841)	(0.2922)
KPMG*Inventory + Receivables	-0.1234	1.1291***	1.3892***
	(0.3246)	(0.3535)	(0.3579)
PwC*Payables	-0.1604	-0.2463	-0.2365
	(0.4643)	(0.4284)	(0.4501)
EY*Payables	-0.2501	-0.3333	-0.2591
	(0.4106)	(0.3164)	(0.3225)
Deloitte*Payables	0.2653	0.1850	0.1778
	(0.3024)	(0.2515)	(0.2529)
KPMG*Payables	-1.1236*	-1.0782*	-1.0683*
	(0.5988)	(0.6117)	(0.6150)
PwC*Ln(Years Client)	1.0348***	0.7803***	0.7140***
	(0.0750)	(0.0785)	(0.0788)

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Table 4 (continued)

EY*Ln(Years Client)	0.8510***	0.6126***	0.5600***
	(0.0641)	(0.0657)	(0.0673)
Deloitte*Ln(Years Client)	0.7122***	0.4582***	0.4226***
	(0.0635)	(0.0654)	(0.0663)
KPMG*Ln(Years Client)	0.8309***	0.6513***	0.6179***
	(0.0625)	(0.0649)	(0.0650)
PwC*Prior Client	2.6462***	2.5439***	2.5421***
	(0.1637)	(0.1628)	(0.1631)
EY*Prior Client	2.0007***	1.8369***	1.8351***
	(0.1391)	(0.1362)	(0.1369)
Deloitte*Prior Client	1.7904***	1.6713***	1.6566***
	(0.1387)	(0.1348)	(0.1353)
KPMG*Prior Client	1.7906***	1.6416***	1.6330***
	(0.1347)	(0.1342)	(0.1338)
PwC*Office in MSA	1.2754***	1.1401***	1.2145***
	(0.2439)	(0.2624)	(0.2631)
EY*Office in MSA	1.5540***	1.4679***	1.4580***
	(0.2039)	(0.2120)	(0.2142)
Deloitte*Office in MSA	0.9657***	0.7309***	0.7778***
	(0.2355)	(0.2709)	(0.2708)
KPMG*Office in MSA	1.7903***	1.3591***	1.3627***
	(0.2318)	(0.2527)	(0.2501)
PwC*Loss	0.0075	-0.0147	-0.0133
	(0.1146)	(0.1148)	(0.1166)
EY*Loss	0.2786***	0.2198**	0.2460**
	(0.0931)	(0.0955)	(0.0960)
Deloitte*Loss	-0.0543	0.1019	0.1101
	(0.0931)	(0.0966)	(0.0971)
KPMG*Loss	0.1503*	0.2043**	0.2134**
	(0.0905)	(0.0947)	(0.0941)
PwC*BTM	-0.0828*	-0.0988**	-0.1022**
	(0.0460)	(0.0449)	(0.0445)
EY*BTM	-0.1757***	-0.1884***	-0.1926***
	(0.0343)	(0.0339)	(0.0339)
Deloitte*BTM	-0.1050**	-0.1102**	-0.1128***
	(0.0411)	(0.0430)	(0.0429)
KPMG*BTM	-0.0174	-0.0111	-0.0198
	(0.0392)	(0.0411)	(0.0411)
PwC*Nondurables	0.0792	0.0453	0.0630
	(0.2560)	(0.2749)	(0.2836)

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Table 4 (continued)

PwC*Durables	-0.0750	-0.1903	-0.1299
	(0.3137)	(0.3486)	(0.3458)
PwC*Manufacturing	-0.1999	-0.0186	-0.1303
	(0.2088)	(0.2122)	(0.2165)
PwC*Energy	-1.5812***	-1.2998***	-1.4980***
	(0.2291)	(0.2300)	(0.2337)
PwC*Technology	0.5007***	0.3543**	0.4737***
	(0.1664)	(0.1700)	(0.1728)
PwC*Telecommunications	-0.0006	-0.1046	-0.0094
	(0.3201)	(0.3107)	(0.3141)
PwC*Utilities	-0.5617	-0.6492	-0.6226
	(0.5438)	(0.5305)	(0.5244)
PwC*Shopping	-0.1301	-0.3149	-0.1363
	(0.2064)	(0.2152)	(0.2199)
PwC*Healthcare	0.2953	0.4714**	0.1591
	(0.1838)	(0.1879)	(0.1930)
EY*Nondurables	0.2114	0.0074	0.0995
	(0.2158)	(0.2288)	(0.2305)
EY*Durables	0.4387	0.0261	0.1611
	(0.2895)	(0.3091)	(0.3127)
EY*Manufacturing	0.3958**	0.1727	0.2838
	(0.1823)	(0.1909)	(0.1926)
EY*Energy	-0.8074***	-0.5346**	-0.5676***
	(0.2058)	(0.2112)	(0.2124)
EY*Technology	0.3010**	0.1565	0.2166
	(0.1442)	(0.1487)	(0.1491)
EY*Telecommunications	-0.8354***	-0.6391**	-0.8276***
	(0.2972)	(0.3040)	(0.3099)
EY*Utilities	-1.4196**	-1.5354***	-1.4000**
	(0.5885)	(0.5841)	(0.5839)
EY*Shopping	0.1186	0.1237	0.0920
	(0.1943)	(0.1996)	(0.2056)
EY*Healthcare	0.9060***	0.7768***	0.7632***
	(0.1478)	(0.1538)	(0.1537)
Deloitte*Nondurables	-0.0789	-0.1611	-0.2319
	(0.2242)	(0.2287)	(0.2373)
Deloitte*Durables	0.0450	-0.0345	0.0253
	(0.2935)	(0.3078)	(0.3081)
Deloitte*Manufacturing	0.2727	0.3076	0.3138*
	(0.1774)	(0.1874)	(0.1896)

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Table 4 (continued)

Deloitte*Energy	-0.7101***	-0.4269**	-0.4019*
	(0.2155)	(0.2139)	(0.2173)
Deloitte*Technology	0.5340***	0.6351***	0.6932***
	(0.1465)	(0.1555)	(0.1566)
Deloitte*Telecommunications	0.0593	0.1591	0.2412
	(0.2626)	(0.2621)	(0.2613)
Deloitte*Utilities	-1.1392**	-0.6388	-1.1557**
	(0.5503)	(0.5372)	(0.5448)
Deloitte*Shopping	-0.1192	0.0627	-0.0119
	(0.1820)	(0.1894)	(0.1938)
Deloitte*Healthcare	0.5044***	0.7109***	0.7858***
	(0.1598)	(0.1650)	(0.1671)
KPMG*Nondurables	-0.1874	-0.3904*	-0.3926
	(0.2214)	(0.2370)	(0.2414)
KPMG*Durables	-0.1564	-0.1251	-0.1888
	(0.3137)	(0.3134)	(0.3133)
KPMG*Manufacturing	0.1297	0.0978	0.0968
	(0.1833)	(0.1922)	(0.1928)
KPMG*Energy	-0.7336***	-0.9930***	-0.9497***
	(0.2145)	(0.2192)	(0.2195)
KPMG*Technology	0.1187	0.2522	0.1603
	(0.1526)	(0.1618)	(0.1620)
KPMG*Telecommunications	-0.3300	-0.4191	-0.3974
	(0.2852)	(0.2943)	(0.2914)
KPMG*Utilities	-1.5009***	-1.4545***	-1.4829***
	(0.5136)	(0.4859)	(0.4856)
KPMG*Shopping	-0.0637	-0.0627	-0.1416
	(0.1810)	(0.1925)	(0.1905)
KPMG*Healthcare	0.4137**	0.4675**	0.5069***
	(0.1675)	(0.1828)	(0.1836)
N	249,557	249,557	249,557
Pseudo R^2	0.8693	0.8785	0.8796
-L	-10498.80	-9759.22	-9676.55
N clust	7,055	7,055	7,055

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Table 4 (continued)

Notes: This table presents estimates of demand over the period 2003-2018. Columns I and II present the results when including industry specialization at the firm and office level separately. Column III presents the results when including both measures simultaneously. We use Column III as our baseline model.
All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.
***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 5: Demand estimation: Interdependencies

Model	I	II	III
ISP_firm	7.5007***	1.2349***	0.6244***
	(0.4685)	(0.0884)	(0.0639)
ISP_office	1.8291***	9.7723***	3.4995***
	(0.0522)	(0.3721)	(0.1271)
ISP_office*ISP_firm	-1.0890***	-0.9546***	
	(0.1109)	(0.1120)	
AO_size	0.3487***	0.3954***	0.1556***
	(0.0200)	(0.0246)	(0.0143)
ISP_firm*AO_size	-0.5735***		
	(0.0432)		
ISP_office*AO_size		-0.7468***	
		(0.0343)	
ISP_office*Big 4			-1.9389***
			(0.1347)
Ln(Audit Fees)	-2.5184***	-2.2936***	-1.0959***
	(0.2224)	(0.2237)	(0.1528)
Residual	-1.4061	-1.2508	-0.6815
	(1.0443)	(1.0551)	(0.8479)
Prior Client	3.0731***	3.0404***	4.0207***
	(0.0640)	(0.0640)	(0.0680)
Controls	Yes	Yes	Yes
N	249,557	249,557	349,469
Pseudo R^2	0.8818	0.8845	0.8724
-L	-9496.91	-9284.16	-12403.33
N clust	7,055	7,055	7,055

Notes: This table presents estimates for demand when including additional interaction terms to test for interdependencies between auditor attributes. In Columns I and II, we interact industry specialization at the office level with industry specialization at the firm level ($ISP_office*ISP_firm$). In Column I, we additionally interact audit office size with industry specialization at the firm level ($ISP_firm * AO_size$). In Column II, we additionally interact audit office size with industry specialization at the office level ($ISP_office * AO_size$). In Columns I and II, the remaining independent variables are the same as in the baseline model of Table (4). In column III, we add two medium-sized audit firms to the choice set to test for the interdependence between industry specialization at the office level and being a Big 4 audit firm ($ISP_office * Big4$). Column III includes all variables as in the baseline model of Table (4) and in addition fixed effects for the two medium-sized audit firms and interaction terms between these additional fixed effects and client characteristics. Table A10 of the Online Appendix shows the complete set of estimates for Column III.

All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 6: Demand estimation: Differential effects

Model	Ln(Assets)	Ln(Segments)
ISP_firm	1.1277***	0.8863***
	(0.3564)	(0.1557)
ISP_firm*attributes	-0.0365	-0.2249
	(0.0276)	(0.1469)
ISP_office	-0.8583***	0.7928***
	(0.2658)	(0.1190)
ISP_office*attributes	0.1914***	0.8386***
	(0.0207)	(0.1166)
AO_size	0.3180***	0.3183***
	(0.0184)	(0.0185)
Ln(Audit Fees)	-2.6433***	-2.6452***
	(0.2239)	(0.2252)
Residual	-1.2058	-1.2558
	(1.0285)	(1.0287)
Prior Client	3.0217***	3.0635***
	(0.0644)	(0.0643)
Controls	Yes	Yes
N	249,557	249,557
Pseudo R^2	0.8802	0.8800
-L	-9624.17	-9640.84
N clust	7,055	7,055

Notes: This table presents estimates for demand when including additional interaction terms to test for differential effects. In column I, we interact industry specialization at the firm level (ISP_firm) and office level (ISP_office) with client size ($Ln(Assets)$). In column II, we interact industry specialization at the firm level (ISP_firm) and office level (ISP_office) with client complexity ($Ln(Segments)$). The remaining independent variables are the same as in the baseline model of Table (4).

All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 7: Robustness test: Defining market leaders as industry specialists

	I	II	III
ISP_firm	0.6340***	6.6296***	1.0501***
	(0.0538)	(0.4463)	(0.0725)
ISP_office	1.4382***	1.7301***	8.9369***
	(0.0470)	(0.0556)	(0.3689)
ISP_office*ISP_firm		-1.0583***	-0.8220***
		(0.1093)	(0.1093)
AO_size	0.3421***	0.3731***	0.3917***
	(0.0184)	(0.0198)	(0.0224)
ISP_firm*AO_size		-0.5073***	
		(0.0411)	
ISP_office*AO_size			-0.6839***
			(0.0344)
Ln(Audit Fees)	-2.8226***	-2.6534***	-2.4769***
	(0.2295)	(0.2265)	(0.2265)
Residual	-1.3467	-1.8092	-1.5094
	(1.0396)	(1.0435)	(1.0544)
Prior Client	3.0660***	3.0397***	3.0036***
	(0.0645)	(0.0642)	(0.0639)
Controls	Yes	Yes	Yes
N	249,557	249,557	249,557
Pseudo R^2	0.8772	0.8792	0.8813
-L	-9870.15	-9705.77	-9539.77
N clust	7,505	7,505	7,505

Notes: This table presents estimates for demand when defining industry specialization based on market leadership. In this table, we define ISP_firm (ISP_office) to be equal to 1 if the audit firm has the largest market share in terms of audit fees at the firm (MSA) level within the SIC2 industry, and 0 otherwise. Column I presents the results of the robustness tests for the baseline model of Table (4). Columns II and III present the results of the robustness tests for interdependencies between auditor attributes of Table (5).

All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 8: Robustness test: Restricting sample to large markets

	I	II	III
ISP_firm	0.6031***	5.8332***	1.0362***
	(0.0657)	(0.5043)	(0.0901)
ISP_office	1.2051***	1.4124***	7.0701***
	(0.0485)	(0.0537)	(0.4093)
ISP_office*ISP_firm		-0.8284***	-0.7916***
		(0.1153)	(0.1160)
AO_size	0.2669***	0.2986***	0.3296***
	(0.0235)	(0.0248)	(0.0270)
ISP_firm*AO_size		-0.4400***	
		(0.0464)	
ISP_office*AO_size			-0.5265***
			(0.0376)
Ln(Audit Fees)	-2.5993***	-2.4857***	-2.3509***
	(0.2410)	(0.2392)	(0.2413)
Residual	-1.1156	-1.2775	-1.2112
	(1.1075)	(1.1121)	(1.1155)
Prior Client	3.0028***	2.9969***	2.9904***
	(0.0682)	(0.0681)	(0.0681)
Controls	Yes	Yes	Yes
N	206,201	206,201	206,201
Pseudo R^2	0.8698	0.8712	0.8722
-L	-8643.88	-8550.82	-8485.15
N clust	6,353	6,353	6,353

Notes: This table presents estimates for demand when excluding all markets at the MSA-industry level with only one client. Column I presents the results of the robustness test for the baseline model of Table (4). Columns II and III present the results of the robustness tests for interdependencies between auditor attributes of Table (5).

All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 9: Robustness test: Restricting sample to clients with auditors of same MSA

	I	II	III
ISP_firm	0.6874***	6.5658***	1.2230***
	(0.0770)	(0.5829)	(0.1079)
ISP_office	1.5192***	1.7476***	9.7580***
	(0.0573)	(0.0632)	(0.4800)
ISP_office*ISP_firm		-1.0441***	-0.9299***
		(0.1342)	(0.1368)
AO_size	0.3070***	0.3334***	0.3757***
	(0.0254)	(0.0267)	(0.0315)
ISP_firm*AO_size		-0.4875***	
		(0.0537)	
ISP_office*AO_size			-0.7497***
			(0.0440)
Ln(Audit Fees)	-2.8632***	-2.7038***	-2.4404***
	(0.3004)	(0.2999)	(0.3000)
Residual	-1.0694	-1.3138	-1.1704
	(1.2643)	(1.2719)	(1.2834)
Prior Client	3.0014***	2.9877***	2.9546***
	(0.0812)	(0.0808)	(0.0810)
Controls	Yes	Yes	Yes
N	157,489	157,489	157,489
Pseudo R^2	0.8720	0.8739	0.8767
-L	-6489.18	-6395.47	-6252.40
N clust	4,566	4,566	4,566

Notes: This table presents estimates for demand when excluding all clients choosing auditors outside their MSA. Column I presents the results of the robustness test for the baseline model of Table (4). Columns II and III present the results of the robustness tests for interdependencies between auditor attributes of Table (5).

All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Table 10: Robustness test: Controlling for information spillover effects

	I	II	III
ISP_firm	0.6962***	7.5420***	1.2482***
	(0.0627)	(0.4681)	(0.0889)
ISP_office	1.5895***	1.8323***	9.8060***
	(0.0472)	(0.0523)	(0.3721)
ISP_office*ISP_firm		-1.0903***	-0.9550***
		(0.1110)	(0.1120)
AO_size	0.3171***	0.3466***	0.3939***
	(0.0186)	(0.0200)	(0.0247)
ISP_firm*AO_size		-0.5761***	
		(0.0432)	
ISP_office*AO_size			-0.7497***
			(0.0343)
Ln(Audit Fees)	-2.6843***	-2.5347***	-2.3097***
	(0.2254)	(0.2232)	(0.2245)
Residual	-1.1022	-1.3280	-1.1548
	(1.0329)	(1.0452)	(1.0553)
Prior Client	3.0876***	3.0761***	3.0428***
	(0.0642)	(0.0642)	(0.0641)
PwC*Herfindahl	-1.0275	-1.1064	-1.4708**
	(0.6818)	(0.6892)	(0.7092)
EY*Herfindahl	-1.2990**	-1.0521*	-1.1887**
	(0.5714)	(0.5707)	(0.5822)
Deloitte*Herfindahl	0.0230	0.1742	0.0771
	(0.5427)	(0.5322)	(0.5489)
KPMG*Herfindahl	-1.1726**	-1.5043***	-1.7269***
	(0.4956)	(0.5109)	(0.5216)
Controls	Yes	Yes	Yes
N	249,557	249,557	249,557
Pseudo R^2	0.8796	0.8819	0.8846
-L	-9671.37	-9490.75	-9276.28
N clust	7,055	7,055	7,055

Notes: This table presents estimates for demand when including interaction terms between auditor fixed effects and the Herfindahl index. We compute the Herfindahl index at the SIC2 level and use it as a proxy for information spillover concerns. Column I presents the results of the robustness test for the baseline model of Table (4). Columns II and III present the results of the robustness tests for interdependencies between auditor attributes of Table (5).

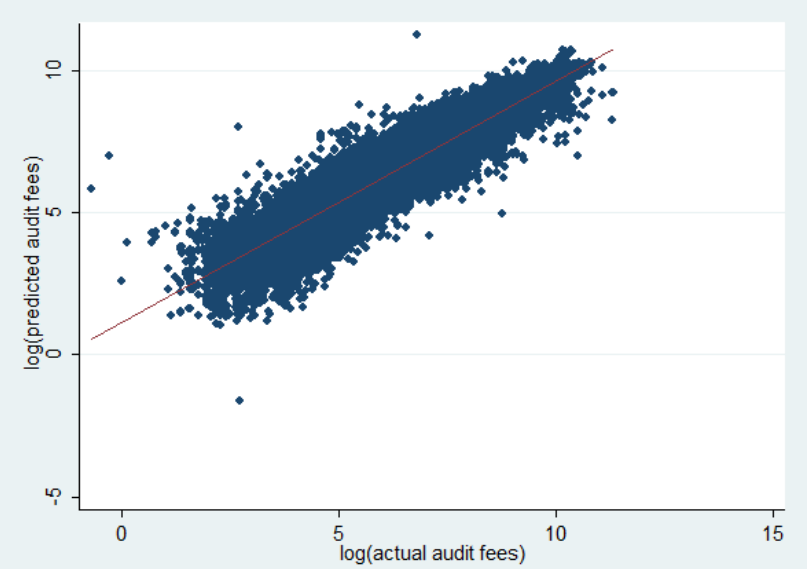
All variables are defined in the Appendix. Standard errors are clustered at the firm level and are reported in brackets.

***, ** and * denote significance at the 1%, 5%, and 10% levels.

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Figures

Figure 1: Actual versus predicted audit fees

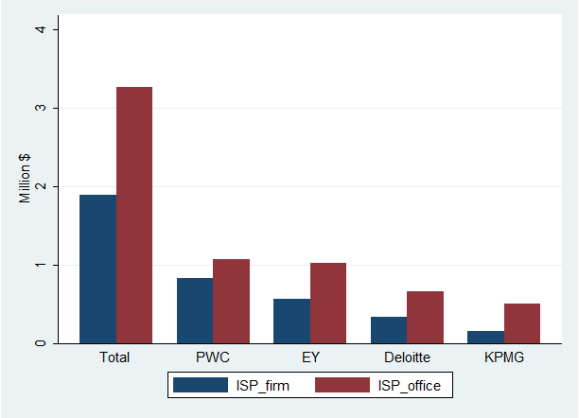


Notes: This figure plots predicted versus actual log audit fees. Predicted fees are generated on an auditor-year basis using an OLS regression using the same sample and the same set of independent variables as in Table (3).

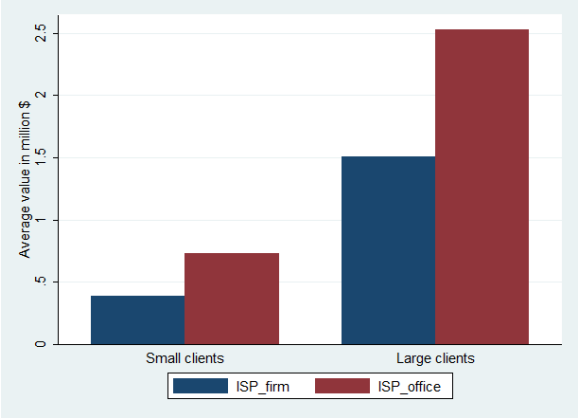
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Figure 2: The value of auditor industry specialization

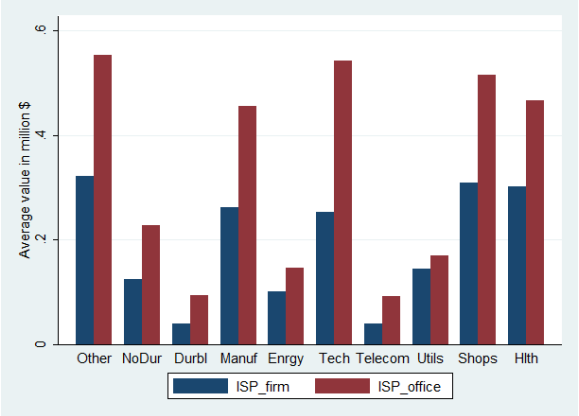
(a) Panel A: Total value and value by audit firm



(b) Panel B: Value conditional on client size



(c) Panel C: Value conditional on client industry

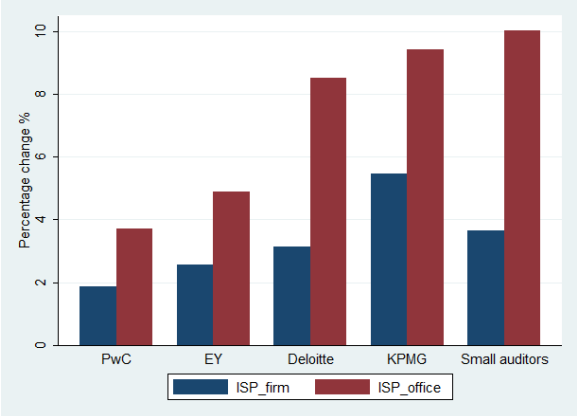


Notes: This figure shows the value of industry specialization. We obtain the value in a counterfactual analysis that restricts clients from hiring industry specialists (Section 5.1.2). Panel A shows the value of industry specialization at the firm level and the office level (total and by audit firm). Panel B shows the value of industry specialization separately for small and large clients. We classify clients into small and large by performing a median split based on total assets. Panel C shows the value of industry specialization by industry based on the Fama-French 10-industry classification.

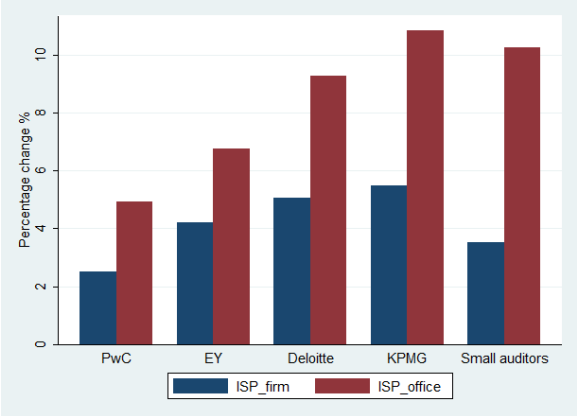
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Figure 3: Market share change when becoming a specialist in all industries (based on number of clients)

(a) Panel A: With fee premium for industry specialization



(b) Panel B: Without fee premium for industry specialization

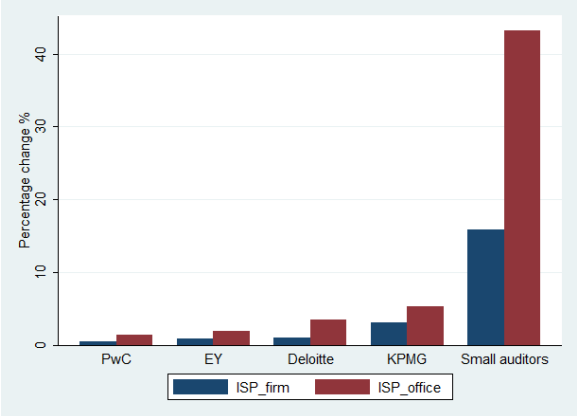


Notes: This figure shows the market share changes in terms of the number of clients when the respective audit firm becomes a specialist in all industries (Section 5.2). Panel A shows the market share changes when considering a fee premium for industry specialization. Panels B shows the market share changes without considering a fee premium for industry specialization.

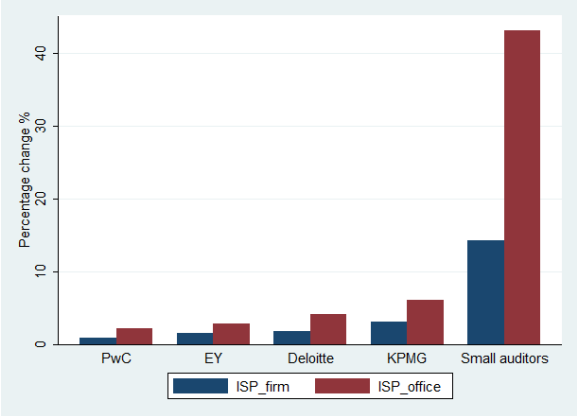
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Figure 4: Market share changes when becoming a specialist in all industries (based on clients' assets)

(a) Panel A: With fee premium for industry specialization



(b) Panel B: Without fee premium for industry specialization



Notes: This figure shows the market share changes in terms of clients' assets when the respective audit firm becomes a specialist in all industries (Section 5.2). Panel A shows the market share changes when considering a fee premium for industry specialization. Panels B shows the market share changes without considering a fee premium for industry specialization.

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Appendix

Appendix: Variable definition

ISP_firm	Firm industry specialist: 1 if the audit firm has a market share of more than 30% in terms of audit fees at the firm level within the industry (SIC2), 0 otherwise
ISP_office	Office industry specialist, 1 if the audit firm has a market share of more than 30% in terms of audit fees at the local level (MSA), 0 otherwise
AO_size	Auditor office size: natural logarithm of total annual revenues of the audit firm in an MSA
MA_SIC3	Instrumental variable: ratio of firms involved in M&A activity in the industry (SIC3) in terms of assets in the previous year
Ln(Assets)	Natural logarithm of total assets (in thousand dollars)
Ln(Segments)	Natural logarithm of the number of business segments
Foreign Sales	Ratio of total revenue from foreign markets to total assets
Debt	Ratio of short-term plus long-term debt to total assets
ROA	Return on assets
Loss	Indicator variable: 1 if net income is negative, 0 otherwise
Inventory + Receivables	Ratio of inventory and receivables to total assets
Payables	Ratio of account payables to total assets
Current Ratio	Ratio of current assets to current liabilities
BTM	Ratio of book value to market value
Ln(Audit Fees)	Natural logarithm of audit fees (in thousand dollars)
Ln(Years Client)	Natural logarithm of the number of years of tenure with the company's incumbent audit firm
Prior Client	Indicator variable: 1 if the company is a client of the audit firm in the prior year, 0 otherwise
Office in MSA	Indicator variable: 1 if the company's incumbent audit firm has an office in the firm's MSA and not all four of the Big 4 audit firms have an office in this MSA, 0 otherwise
Herfindahl	Herfindahl concentration index within an industry (SIC2) based on clients' sales (in thousand dollars)

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